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Isabel-María García Sánchez Luis Rodríguez Domínguez Isabel Gallego Álvarez

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# Corporate governance and strategic information on the internet

## A study of Spanish listed companies

Isabel-María García Sánchez, Luis Rodríguez Domínguez and Isabel Gallego Álvarez

*Facultad de Economía y Empresa, Campus Miguel de Unamuno, Edificio FES, Salamanca, Spain*

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### Abstract

**Purpose** – The purpose of this study is twofold: to evidence the disclosure practices of Spanish companies in relation to a voluntary typology of strategic information; and to determine the factors that explain these practices. Among the factors considered, the study seeks to focus on the role of the Board of Directors in depth. According to Agency Theory, strategic information has positive consequences on external funds costs. On the other hand, Proprietary Costs theory limits these practices, given that they can lead to competitive disadvantages.

**Design/methodology/approach** – First, online strategic information disclosure practices are analysed by examining non-financial quoted Spanish firms. A disclosure index is created, and subsequently, certain factors related to corporate governance – Activity, Size and Board Independence – as well as other factors traditionally analysed, are used to explain the volume of strategic information disclosed on the internet.

**Findings** – The results indicate that Spanish companies, on average, give out little strategic information, mainly related to objectives, their mission, and the company's philosophy. "Company annual planning" and "Information on risks" are scarcely disclosed. The findings also emphasise that companies where the Chairperson of the Board is the same person as the CEO and, moreover, in which there is a lower frequency of meetings, disclose a greater amount of strategic information on their web sites.

**Practical implications** – The findings suggest that the disclosure of strategic information is a decision taken by executives with the aim of satisfying the demands of creditors and investors. The Board of Directors represents the shareholders' interests, but it does not participate in strategic decision-making disclosure, maybe due to the fact that the proprietary costs lack influence.

**Originality/value** – The link between corporate governance and strategic information disclosed online has scarcely been analysed in previous literature. This study provides interesting insights into how several Board characteristics can affect the disclosure of strategic information on the internet.

**Keywords** Corporate governance, Disclosure, Boards of Directors, Internet, Generation and dissemination of information, Spain

**Paper type** Research paper

## 1. Introduction

The voluntary disclosure of strategic information[1] is gradually becoming a more common corporate practice, owing to the benefits to which it leads, such as its ability to make a company stand out from other corporations (Kohut and Segars, 1992; Santema *et al.*, 2005) and its usefulness in the evaluation processes performed by professional

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investors, banks, analysts and financial intermediaries (Higgings and Diffenbach, 1985; AICPA, 1994). In this sense, Holland (1998, p. 65) argues that voluntary strategic information was very useful for corporate financing –by reducing external funds costs– and for control decisions – by preventing managers from using budgetary discretion to pursue their own interests. While Boards of Directors – one of the corporate governance control mechanisms – have internal access to this type of information, an improved level of disclosure of future information to current and prospective investors can help them make informed decisions (Cianci and Falsetta, 2008). Moreover, Good Corporate Governance Codes generally recommend that Boards of Directors encourage the disclosure of information that can affect market prices (Lim *et al.*, 2007, p. 556) –such as concerning financial and management risks or innovation activities. Furthermore, recent financial scandals have led to a demand for increased transparency in corporate reporting, in order to restore confidence in the current context.

On the other hand, apart from the benefits[2], it can lead to significant competitive disadvantages. This could mean that the factors explaining the decision to disclose such information will potentially differ from those that explain decisions leading to the publication of other kinds of information (Lim *et al.*, 2007).

Previous studies have found a high degree of similarity in the contents of the strategic information disclosed[3], when comparing quoted and unquoted companies and amongst different types of firms, according to their product's consumers (Santema and Van de Rijt, 2001). Likewise, the geographic location (Santema *et al.*, 2005), listing status in a foreign market (Gray *et al.*, 1995) and independence of the Board (Lim *et al.*, 2007) contribute to explaining some significant differences in strategic information disclosure practices.

In relation to the last factor – the Board – several papers have stated its significant role in the strategic decision-making process (e.g. Pearce and Zahra, 1992; Dalton *et al.*, 1999; Chatterjee *et al.*, 2003) and in the disclosure of compulsory and voluntary information (e.g. Chen and Jaggi, 2000; Willekens *et al.*, 2005; Cheng and Courtenay, 2006), which is why it could be of interest to extend the proposal by Lim *et al.* (2007) to examine other features of corporate governance effectiveness –mainly, Board independence, activity and size – in order to establish their weight in the corporate decision to disclose strategic information online.

In light of the above arguments, the objective of this paper is twofold: first of all, to analyse online strategic information disclosure practices by examining non-financial quoted Spanish firms as the study set; second, as the main and most innovative objective, to establish the factors that explain the volume of strategic information disclosed on the internet. Specifically, we analyse the impact of the Board size, activity and independence on the online disclosure of strategic information through some hypotheses included within several theories (Agency, Signalling, Political Costs and Proprietary Costs Theories) and taking into consideration the confirmed effect of the members of the Board on the disclosure of compulsory, voluntary (e.g. Chen and Jaggi, 2000; Willekens *et al.*, 2005; Cheng and Courtenay, 2006) and strategic (Lim *et al.*, 2007) information through the traditional channels for releasing annual reports. Moreover, we controlled for several variables, such as size, leverage, profitability and industry.

We take the volume of information disclosed as a proxy for substantive disclosure. Undoubtedly, this approach involves some limitations in our study. However, we

assume that this volume and the indicators displayed should also be qualitative in order to be informative.

We have selected the Spanish context for two key reasons:

(1) the existence of a requirement for Spanish listed companies to provide relevant information on web pages according to Circular 1/2004 of the Comisión Nacional del Mercado de Valores (*CNMV*, 2004); and

(2) the interest in extending the empirical previous evidence, which is generally from Anglo-Saxon settings.

In this sense, this country presents several differences that generate a higher interest in analysing the role of the Board on firms' disclosure practices. First, the Spanish economy is characterised by a low proportion of listed companies compared to the Anglo-Saxon markets. Second, the stock ownership is highly concentrated in the hands of non-financial companies, financial institutions and families[4]. Third, Boards of Directors in Spain – as occurs in most European countries – are one-tiered, which means that Board members manage the company and also supervise its activity (Melle, 1999), thus promoting an active participation in the taking of strategic decisions. Fourth, the legal protection of shareholders is not as extensive as that of Anglo-Saxon markets and Spanish stock markets are less developed and play a far lesser role than British or American markets do (Fernández-Méndez and Arrondo-García, 2007). In this sense, it is necessary to disclose more information, especially of the strategic type, in order to improve the shareholders' knowledge and trust in the firm's behaviour and performance.

Our results suggest that upper-level management plays an important role in the decision to disclose strategic information online.

This paper is organised as follows: Section 2 discusses information disclosure, emphasising its advantages and disadvantages, the importance of strategic information and the role played by the internet as a mechanism for disclosure; section 3 deals with corporate governance, by describing its role in information disclosure; section 4 describes our hypotheses; section 5 describes the research design, including the creation of a disclosure index, the proposed model and the sample used. Our empirical results are reported in section 6, and we provide a summary and conclusion in section 7.

## 2. The disclosure of information

### *2.1 Advantages and disadvantages of disclosing information*

To disclose information is one of the most important decisions made by corporations, because of its potential consequences, and the advantages and disadvantages that stem from doing so. In terms of the advantages, from an Agency Theory perspective, the information that is provided can be useful in the decision-making process of owners and managers, and it can work as a system for control by shareholders and other stakeholders over manager activities (Jensen and Meckling, 1976). According to Signalling Theory, information disclosure may be regarded as a signal to capital markets, so as to decrease information asymmetries, optimise financing costs[5] and increase corporate value (Baiman and Verrecchia, 1996). In accordance with Political Costs Theory, companies voluntarily disclose information that might lead to regulation orientated towards decreasing political costs (e.g. taxes) and obtaining several

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advantages (e.g. subsidies, government action that favours the corporation). Other advantages related to information disclosure are associated with an improvement in corporate image, an increase in investor trust (Babío *et al.*, 2003), greater stock liquidity (Healy *et al.*, 1999; Guo *et al.*, 2004) and obtaining larger volumes of funds (Marr and Gray, 2002).

Nevertheless, an optimal level of voluntary disclosures involves a trade-off between the benefits and cost of revealing proprietary information (Hayes and Lundholm, 1996). Producing and disclosing information voluntarily may also lead to disadvantages that may sometimes outweigh the benefits. For instance, the costs linked to a likely competitive disadvantage and political costs are widely argued as a justification for the reluctance of managers to provide more information than that legally required (Gray *et al.*, 1990; Guo *et al.*, 2004).

In this sense, Proprietary Costs Theory argues that there are two types of costs associated with information disclosure: direct and indirect costs. Direct costs are those that have to do with the processing, collection and dissemination of information. They do not generally justify the fact that a company does not carry out the disclosure of information in terms of costs, because this information must be necessarily prepared for making internal decisions. Further these costs have arguably fallen with the emergence of online reporting (Gallhofer and Haslam, 2007).

However, the most relevant drawback to disclosing information voluntarily stems from the potential damage derived from the use of the information for the firm. In this vein, it is worth emphasising the costs derived from competitive damage, since the information would be public not only for current and potential investors, but for competitors, who could detect possible opportunities to improve their market positions. In addition to the danger of providing useful information to both existing and potential competitors, voluntary disclosures may increase the political costs, both direct and indirect, faced by corporations if disclosures reveal, for example, situations of monopolistic advantage or social inequalities. There are also other potential disadvantages derived from voluntary disclosures (Gray *et al.*, 1990): threats of takeovers or mergers, possibilities of intervention by government agencies and taxation authorities, and possibilities of claims from employees or trade unions or from political or consumer groups. In this vein, the information would be also available for other interested users, such as governments, trade unions, consumer associations, clients or suppliers, thereby leading to likely increases in pressures on the firm (e.g. demands related to prices or salaries).

Moreover, the disclosure of the information could lead to costs derived from litigations companies must face as a consequence of lawsuits undertaken by stockholders and other stakeholders, due to the damage generated by the information and the generation of expectations that in the end are not fulfilled.

Likewise, by disclosing strategic plans, managers become reluctant to change their minds in the future and this reluctance may lead them to make inefficient project implementation decisions (Ferreira and Rezende, 2007). At the same time, managers may try to avoid setting disclosure precedents that will be difficult to maintain in the future (Graham *et al.*, 2005).

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*2.2 The disclosure of strategic information*

The most notable information of a non-financial nature –mainly information not related to financial statements– that companies currently divulge, because of its relation to the company's future, is strategic information (Lim *et al.*, 2007).

At first, research suggests that companies began to include their missions or visions as a part of their annual reports. In general, they were included with no assurance by managers that the contents of this information explicitly coincided with the firm's true mission (Bart, 1997). The content was mainly provided as part of a policy oriented towards "standing out". More specifically, the ultimate goal was to identify the benefits or advantages the company provided in comparison with its competitors, as well as its business values, attitudes and norms, and it was intended for, in this order of priority, customers, investors, employees and suppliers (Leuthesser and Kohli, 1997; Campbell *et al.*, 2001). On occasion, the objective was also to determine the fulfilment of strategic goals in prior years (Santema and Van de Rijt, 2001).

Despite many studies that find strategic information to be useful when professional investors and analysts evaluate companies (Higgings and Diffenbach, 1985; AICPA, 1994), there is still a lack of information. This situation may arise because corporations are unaware of the benefits involved in disclosing this kind of information (Bart, 1997), the fear of possible lawsuits as a result of failing to fulfil strategic objectives (Santema and Van de Rijt, 2001) and confidentiality problems (Vacas *et al.*, 2005). Moreover, future information is not necessarily always positive and, in this sense, can affect corporate value negatively (Skinner, 1994, p. 38).

Currently, the disclosure of strategic information is heterogeneous when we take the company's geographic location into consideration (Santema *et al.*, 2005). However, there is a high level of consistency in the contents of strategic information over the time periods analysed; also, there is a high level of similarity in the strategic information disclosed between quoted and unquoted companies in the same country (Santema and Van de Rijt, 2001). There is a positive influence of a listing status in international markets on the volume of disclosure (Gray *et al.*, 1995); at the same time, the presence of independent Board members on the Board or some entity responsible for monitoring the company positively influences the disclosure of that type of corporate information (Lim *et al.*, 2007).

Nowadays, as Bartkus *et al.* (2002) have shown, companies – mainly European firms – have begun to reveal this sort of information on their web sites, owing to the numerous advantages the internet provides (Gandía and Andrés, 2005).

*2.3 Disclosing information on the internet*

Though relatively recent, the disclosure of online information has undergone dramatic growth according to several studies, such as Petravick and Guillet (1996, 1998), Gray and Debreceny (1997) and Debreceny *et al.* (1999) in the US; Lymer (1998) and Craven and Martson (1999) in the UK; and Hedlin (1999) in Sweden. In Spain, however, Gowthorpe and Amat (1999) find something of a delay compared to the international scene.

The advantages to a corporation of supplying information on a company web site include providing individual investors with a quantity and timeliness of information previously available only to select parties, such as institutional investors and analysts. Compared with traditional printed reports, the internet offers many more opportunities

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to communicate corporate information and allows a wealth of up-to-date, unofficial, critical and alternative channels of accounting information to compete with the official channel (Paisey and Paisey, 2006; Gallhofer *et al.*, 2006).

Corporations now have the ability to deliver unfiltered information to their publics and without a time lag. Internet communication is multidirectional in nature and very fast in transmission. Easy access to the new information technology, such as the web site, now empowers individuals to easily disseminate their viewpoints on anything. In addition, some corporate costs of printing or mailing may be reduced, at the same time that investors can obtain information at less cost. The internet can also create interest among potential investors and provide a boost to the corporate image (Noack, 1997). This medium allows the company to control the context in which data are presented, emphasise the positive aspects and provide interpretation for potentially negative information.

Consequently, the internet can be an appropriate and useful way of communicating corporate strategic information[6]. Likewise, it is possibly the most powerful means of providing targeted information to specific concerned stakeholders as a legitimation strategy (Campbell *et al.*, 2003).

Empirical evidence has made it clear that large-sized companies (Ashbaugh *et al.*, 1999; Craven and Martson, 1999; Pirchegger and Wagenhofer, 1999; Oyelere *et al.*, 2003; Marston and Polei, 2004; Serrano-Cinca *et al.*, 2007) use the internet as a channel of communication with their stakeholders more frequently than smaller firms do. Moreover, there is relative homogeneity in the information disclosed in certain sectors of activity (Debreceny *et al.*, 2002; Oyelere *et al.*, 2003; Bonsón and Escobar, 2004; Xiao *et al.*, 2004).

## 3. The active role of corporate governance

### *3.1 Corporate governance and strategy*

An Agency Theory approach here is used descriptively for a number of reasons. First, this facilitates comparison with results from previous studies –, e.g. the work of Lim *et al.* (2007) for Australian corporations and Depoers (2000) for French public companies, which used the agency framework to analyse the role of the Board in strategic information disclosure practices. The second motive is the evidence of previous studies, which observed that voluntary disclosure information is not determined by stakeholders' demands (Gray *et al.*, 1990; Meek *et al.*, 1995), but it is regarded as a trade-off between perceived proprietary costs that could damage shareholders' interests and a mitigation of agency conflicts (Depoers, 2000; Daurus *et al.*, 2008), pointing to the explanatory power of an Agency Theory approach.

Third, the Agency Theory is the most appropriate one for explaining corporate behaviour in Spain. For instance, the information disclosed online is compulsory for Spanish public firms according to Circular 1/2004 of the Comisión Nacional del Mercado de Valores (*CNMV*, 2004). The rule states that this information must be contained in a section addressed to stockholders, reflecting Agency Theory's arguments- instead of a stakeholder accountability model. In this respect, previous Spanish evidence of the role played by corporate governance in voluntary disclosure (e.g. Gallego-Álvarez *et al.*, 2008, 2009; Rodriguez-Dominguez *et al.*, 2009a), firm performance (e.g. García-Sánchez, 2010a) and the presence of ethics codes (e.g. García

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Sánchez *et al.*, 2008; Rodríguez Domínguez *et al.*, 2009b) supports conclusions according to Agency Theory postulates.

Likewise, in every firm, a clear and precise strategy is an element of vital importance (Ansoff, 1957; Porter, 1980). In this vein, Charkham (1994) argues that the starting point for the taking of strategic decisions must be framed in the context of corporate governance, which analyses the process of monitoring decisions and actions, as well as the capability of influencing them. Zahra (1990) establishes three levels of the implication of Boards in strategic decisions. At the first level – the most inactive – the function of Boards is to represent and protect stockholders' interests. This is achieved without becoming involved in the development and implementation of strategies, except when the performance of management is extremely adverse. At the second level, Boards are quite active – both in formulation and implementation of corporate strategies – but only in ratification or rejection of the strategies proposed by top management. At the third level, Boards develop their monitoring functions to participate actively in the corporate strategy. In relation to this third level, Mizruchi (1983) argues that Boards set the parameters within which the strategy process takes place, since a more active participation by the Board in implementing the strategy may lead to significant improvements in returns.

Previous studies attempting to analyse the effect of corporate governance on the strategic decision-making process, have found a positive relationship. For instance, Pearce and Zahra (1992) and Dalton *et al.* (1999) researched the effect of the size of the Board on the process for planning new strategies, whereas Chatterjee *et al.* (2003) determined that there is a strong link between the Board's independence – measured as the percentage of external board members – and the control exerted over the strategy. We may claim that an active Board controls the upper-level management and influences its decisions, by giving advice on the activities it carries out (Johnson *et al.*, 1996).

### *3.2 Corporate governance and information disclosure*

Per Agency Theory, the disclosure of information is a decision reached by the Board in order to reduce information asymmetry and agency costs (Lev, 1992; Richardson and Welker, 2001). This process leads to an increase in transparency and, consequently, a reduction in firm's capital cost, at the same time enabling enhancement of the organisation's status and reputation.

Jensen and Meckling (1976) provide a framework of analysis in which they establish a likely complementary or substitutive link between disclosure practices and internal mechanisms of corporate governance. Therefore, depending on the character of the link, the effectiveness of the corporate governance would cause a higher or lower volume of information to be disclosed to shareholders.

Previous research has not reached clear results on this type of link. Forker (1992) and Ho and Wong (2001) found a lack of any relationship between corporate governance and the disclosure of compulsory information and voluntary information, respectively. Eng and Mak (2003) found a negative relationship between voluntary disclosure and the presence of independent Board members, results that were confirmed in Gul and Leung (2004)'s work, in which the negative effect of independent directors' reputation for disclosing this kind of information was found.

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Nevertheless, other studies have confirmed the complementary link, by verifying that the effectiveness of the Board is associated with a higher amount of financial information disclosed (Chen and Jaggi, 2000; Willekens *et al.*, 2005), with its level of quality (Karamanou and Vafeas, 2005), and with voluntary information (Cheng and Courtenay, 2006); however, these findings are not supported for non-financial and historical information (Lim *et al.*, 2007) or for information on broader corporate social responsibility (Haniffa and Cooke, 2005).

## 4. Hypotheses

The following are the main research hypotheses related to several factors that may influence the amount of information that is disclosed. These hypotheses will subsequently be researched in the empirical study.

Following the hypothesis that an active Board monitors top management and influences their decisions (Johnson *et al.*, 1996), as a consequence of the benefits owing to this involvement (Zahra, 1990), it is a good idea to take into consideration some parameters of Boards of Directors that may influence their control capability over management and their participation in the decision-making process: activity, size and independence. Building upon prior literature, we also explore other key factors of relevance.

### *4.1 Board activity*

Board activity is generally linked to the Board agenda that typically focuses the work of the Board (Inglis and Weaver, 2000), and which, through the Board meetings, provides a meaningful forum of communication between the directors and managers pertaining to a range of issues spanning from operational reviews to business cycle plans.

The most active Boards of Directors – typically understood as those that meet most frequently – fulfil their duties in accordance with shareholder interests (Conger *et al.*, 1998), because they devote more time to consulting, implementing the corporate strategy and monitoring the upper-level management (Reyes-Recio, 2000). They tend to be more effective, which leads to a higher incentive to disclose more information, allowing stakeholders to become aware of their efforts (Lipton and Lorsch, 1992). For the Spanish context, the Board activity is even more important since it is necessary to reduce asymmetric information problems among different types of directors. In this regard, García-Sánchez (2010b) has observed that those boards and committees with higher executive presence increase their meeting frequency in order to improve the monitoring of the insiders in relation to the quality of financial statements, involving a larger flow of information.

According to Agency Theory arguments, Board activity is expected to have a positive effect on strategic disclosure. There is, however, the issue (also from the shareholder value perspective of relevance in Spain) of protecting proprietary interests, suggesting limiting disclosure that could cause competitive disadvantages, litigation risks or stock price response to bad news. In this sense, an effective Board could decide to: oppose disclosing important information (Prado Lorenzo and García Sánchez, 2009); or reveal a high volume of useful information, as Banghoj and Plenborg (2008) have observed. Therefore, the results of our analysis could indicate whether the expected sign may be questionable.

Considering the scanty empirical evidence available regarding its impact on disclosing information online, we attempted to verify the following hypothesis:

*H1.* The meeting frequency of the Board of Directors influences the amount of strategic information disclosed on web sites.

The variable used to represent the meeting frequency of the Board of activity was the number of meetings held during the year 2005.

*4.2 Board size*

The impact of Board size on its effectiveness is a controversial issue. Several authors affirm that an increase in Board size will lead to the incorporation of Directors with new perspectives to analyse the issues to be dealt with, thereby bringing higher quality to corporate decisions (Pearce and Zahra, 1992). In this line, Pearce and Zahra (1992) and Dalton *et al.* (1999) found that the size of the Board was positively related to the processes for planning new strategies. In this vein, the higher the Board size the higher the volume of strategic information disclosed in order to show their significant efforts. Nonetheless, in accordance with the arguments related to last hypothesis, directors may be specially reluctant to reveal a high volume of strategic information.

Other papers, e.g. Yermack (1996), Eisenberg *et al.* (1998) and Andres *et al.* (2005), have found that the presence of a higher number of Board members would lead to less effectiveness of the Board in terms of management control, as a result of increased agency problems and a likely decrease in Board agility and reaction capability.

In our opinion, this decrease in effectiveness could lead to a lesser predisposition towards revealing information on corporate activities, because of the lack of an appropriate control mechanism, the need for concealing bad news (or non-optimal news) from stockholders or the desire to keep the internal discussions of the Board out of the public's view.

Taking into consideration the above contradictory arguments we can establish the following research hypothesis:

*H2.* A larger Board of Directors influences the volume of strategic information disclosed on web sites.

To represent the size of the Board, we have used the number of Board members.

*4.3 Independence of the Board of Directors*

An independent Board is viewed as a crucial mechanism for monitoring managers' activities and assuring the achievement of stockholders' objectives (Fama and Jensen, 1983; Agrawal and Knoeber, 1996), and for participating in and controlling strategic decisions (Chatterjee *et al.*, 2003). Its level of independence is usually associated with the number of outside directors on the Board, and non-CEO duality (e.g. the CEO is not a Board member).

*4.3.1 Outside Board members' level of involvement.* Outside Board members are members of the Board who are more interested in demonstrating corporate behaviour and fulfilling the established objectives, by representing the shareholders' interests (Lim *et al.*, 2007) and exerting greater control over corporate strategy (Chatterjee *et al.*, 2003).

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With regard to information disclosure, previous evidence has not reached clear results on this relationship. Forker (1992), Ho and Wong (2001), Eng and Mak (2003), Haniffa and Cooke (2005) and Boesso and Kumar (2007) found a negative or non-significant relationship between information disclosure and the presence of independent directors on the Board. Gul and Leung (2004) reached the same conclusions, verifying the negative effect of independent directors' reputations for disclosing the same type of information.

In this sense, and in accordance with Bathala and Rao (1995) and Baysinger and Hoskisson (1990), it is necessary to consider that outside directors may not be able to exert sufficient influence, partially because they lack the superior information possessed by inside directors and partially because of time constraints as a result of multi-firm independent outside director appointments. Likewise, independent directors can sometimes be reluctant to disclose relevant voluntary information that could provoke litigation risks for companies (Prado Lorenzo and García Sánchez, 2009; Prado Lorenzo *et al.*, 2009).

However, other studies have confirmed a positive relationship (Chen and Jaggi, 2000; Karamanou and Vafeas, 2005; Cheng and Courtenay, 2006; Willekens *et al.*, 2005; Lim *et al.*, 2007).

*4.3.2 CEO duality.* Furthermore, CEO duality – the situation in which the Chairperson of the Board holds the managerial position of CEO – usually leads to a decrease in the independence and effectiveness of the Board (Jensen, 1993). This decrease in independence can have negative repercussions on the disclosure of corporate information, as a result of the increase in the power of managers, whose objectives can be the opposite of shareholders". While duality can ensure the unity of command within an organisation, it can also encourage excessive centralisation and limit the information processing capabilities of the board (Zahra *et al.*, 2000, p. 947).

Some researchers emphasise stewardship in this regard. Thus, Donaldson and Davis (1991, p. 51) affirm that, "the executive manager, under this theory, far from being an opportunistic shirker, essentially wants to do a good job, to be a good steward of the corporate assets". Under this paradigm, duality can entail certain advantages associated with the unification of leadership and a great knowledge of the firm's operating environment that should impact positively on firm strategy (Finkelstein and D'Aveni, 1994). More specifically, duality helps enhance decision making by permitting a sharper focus on company objectives and promoting more rapid implementation of operational decisions (Stewart, 1991) and shapes the destiny of the firm with minimal Board interference, which could also lead to improved performance resulting from clear, unfettered leadership of the Board (Rechner and Dalton, 1991).

Nevertheless, we must point out that there is a significant difference between Board composition in Spanish and Anglo-Saxon firms. The weight of non-executive directors reported in the 2003 European survey conducted by Heidrick & Struggles (2003) is 46 per cent for the Spanish listed companies. However, the proportion of outside directors in British and S&P 500 US firms is 90 per cent and 80 per cent as reported, respectively, in Heidrick and Struggles' (2003) survey and the Spencer Stuart (2003) Board index. Moreover, Spanish Boards are characterised by the strong power of executives through CEO duality. So it is possible that outsiders do not participate enough in strategic decision-making due to their limited presence.

The divergence of results for the effect that the presence of independent Board members has on voluntary disclosure joined to the theoretical arguments in favour, we decided to test the following hypothesis:

*H3.* Greater independence of Boards of Directors influences the volume of strategic information disclosed on web sites.

The independence of the Board is measured by: a numeric variable that reflects the percentage of outside Board members on the Board; and a dummy variable that takes a value of 1 if the Chairperson of the Board holds the main managerial position in the company, and 0 otherwise.

*4.4 Blockholders*

Several studies have found that a presence of shareholder representatives leads to the creation of higher quality information (e.g. Beasley, 1996). Owing to the alignment of their objectives with those of the other shareholders, the presence of these Board members may reflect an interest in revealing a larger amount of information, which allows for better control of manager activities.

In Spanish firms, the stock ownership is highly concentrated in the hands of non-financial companies, financial institutions and families in which the supervisory role of large shareholders is quite important.

All these arguments lead us to establish the following research hypothesis:

*H4.* The presence of shareholder representatives on Boards positively influences the volume of strategic information disclosed on web sites.

To check this hypothesis, we use a dummy variable, which takes a value of 1 when there is a shareholder representative on the Board and 0, otherwise.

*4.5 Other explanatory factors*

*4.5.1 Corporate size.* Large-sized companies, according to Agency Theory, present greater needs for external funds and, in consequence, they should disclose a higher volume of voluntary information as a way of diminishing the financial spending, thereby allowing the company to access external funds more competitively. Moreover, larger companies suffer from more conflicts of interest between shareholders, debt-holders and managers. In this situation, it is possible to use the disclosure of voluntary information as one of the ways to reduce information asymmetries.

From a cost-benefit analysis perspective, the costs of preparing and disseminating information on the internet are likely to be independent of corporate size (Larrán and Giner, 2002; Bonsón and Escobar, 2004). Nevertheless, the potential benefits will be greater for larger corporations, because there is a direct relationship between agency costs and disclosure benefits, in the sense that large-sized companies will face greater agency costs and will be more benefited from a higher volume of disclosure. In relation to this argument, larger companies are more visible in markets and in society as a whole, with greater coverage by analysts and greater sensitivity to public image. These users would end up producing a greater demand for information and put pressure on the company to disclose it.

Taking into account these arguments, most previous research has found a positive influence of corporate size on the amount of voluntary information disclosed on web

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sites (Craven and Martson, 1999; Oyelere *et al.*, 2003; Marston and Polei, 2004; Giner *et al.*, 2003; Gul and Leung, 2004; Boesso and Kumar, 2007). Less frequently, there have been other studies finding exceptions to the direct relationship, in the sense that it is valid only up to a certain size (Pirchegger and Wagenhofer, 1999), whereas several works do not find a statistically significant relationship, such as Khanna *et al.* (2004) or Ortiz and Clavel (2006), for European multinationals listed on the NYSE.

As the variable related to corporate size, we selected the market capitalisation in December 2005.

*4.5.2 Industrial sector.* Industry has been one of the variables often used to explain the quantity of information provided by corporations. Companies that do business in the same industry are believed to adopt similar guidelines on the information they disclose. They face the same level of business complexity and industry instability and volatility (Boesso and Kumar, 2007). If a company fails to adopt the same disclosure strategy as other corporations in the same industry, the market could interpret this as bad news (Watts and Zimmerman, 1986, p. 239). Likewise, industry membership may affect the political vulnerability of firms, and therefore companies in industries that are more politically vulnerable may use voluntary disclosure to minimise political costs, such as regulation, or the break-up of the entity/industry (Oyelere *et al.*, 2003).

The results obtained in the previous literature are far from reaching a clear conclusion, as might be the case with corporate size. While some works have found that industry membership contributes to explaining the amount of voluntary information disclosed (Oyelere *et al.*, 2003; Gul and Leung, 2004; Bonsón and Escobar, 2004), especially in the information technology sector or in high growth industries (Xiao *et al.*, 2004), other studies have not shown a statistically significant relationship (e.g. Giner, 1997; Craven and Martson, 1999; Larrán and Giner, 2002; Giner *et al.*, 2003).

In order to analyse the effect of industry we used the CNMV industry classification, including five dummy variables that represent Services, Transportation, Industry, Energy and Construction.

*4.5.3 Profitability.* The link between profitability and voluntary disclosure is especially complex. The main disclosure theories tend to indicate that there is a positive relationship. In accordance with Agency Theory, the managers of profitable companies use information to obtain personal advantages, such as ensuring the stability of their positions and increasing their levels of compensation.

From the perspective of Signalling Theory, profitability can be considered an indicator of the quality of the investment. Therefore, if a high level of profitability is achieved, there will be a greater incentive to disclose information and reduce the risk of being viewed negatively by markets. According to this theory, profitable companies reveal information in order to stand out from other less successful corporations, obtain funds at the lowest cost and avoid any decrease in stock prices.

In addition, Political Costs Theory supports the disclosure of voluntary information, so as to justify the returns obtained.

Nevertheless, Wagenhofer (1990), Giner *et al.* (2003) and Prencipe (2004) have analysed a likely negative relationship, since higher profitability could spur rival companies to enter into the company's market. Consequently, it is essential to consider the influence of competitive costs, which tend to increase when profitability increases.

Despite the coherence of the assumptions of disclosure theories, most previous studies do not find a statistically significant relationship between voluntary disclosure

and profitability (Larrán and Giner, 2002; Oyelere *et al.*, 2003; Giner *et al.*, 2003; Marston and Polei, 2004; Prencipe, 2004; Magness, 2006). Unlike these works, Khanna *et al.* (2004) and Gul and Leung (2004) find a positive influence of profitability on the amount of voluntary disclosure, both in the multinationals listed on the NYSE and in quoted companies in Hong Kong, respectively.

As the variable related to profitability, we have used the return on assets (ROA) in December 2005.

*4.5.4 Leverage.* The amount of leverage is another factor associated with a larger amount of disclosed information, especially as a result of agency conflicts that may arise. In this sense, the companies with more debt have greater agency costs, because there is a possibility of transference of wealth from debt-holders to stockholders. By increasing the amount of information disclosed, corporations can reduce their agency costs and conflicts of interest between owners and creditors.

Moreover, as leverage increases, the demand for additional information requested by creditors also rises, because they will attempt to find out how likely the company is to meet its financial obligations. In terms of stockholders, voluntary information is a mechanism used to monitor management and evaluate the company's financial health, given that the risk of financial distress increases with rising leverage.

In this respect, several studies have found a positive effect of leverage on the amount of information revealed voluntarily (for example, Giner *et al.*, 2003; Xiao *et al.*, 2004; Prencipe, 2004; Alvarez, 2007), whereas other works do not find a statistically significant relationship (Giner, 1997; Oyelere *et al.*, 2003; Gul and Leung, 2004).

In this research paper, we have measured leverage as the debt/total assets ratio at December 2005.

*4.5.5 Ownership diffusion.* In corporate contexts with high ownership dispersion – in other words, where there are many small stockholders who do not directly take part in the company's management or control – the agency costs that stem from information asymmetry are especially high. Faced with these situations, the disclosure of information is viewed as the most appropriate mechanism for reducing such asymmetries between managers and stockholders.

Authors such as McKinnon and Dalimunthe (1993), Saada (1998), Leuz (1999) and Prencipe (2004) have analysed this relationship, with different levels of statistical significance.

To represent this effect, the percentage of shares not owned by the blockholders and by the executives and directors. More concretely, this paper considers the shareholders with a stock ownership higher than 5 per cent as blockholders.

## 5. Research methods

### *5.1 Sample description*

In order to research the hypotheses, we used a sample of companies listed on the Madrid Stock Market. The initial sample was made up of all the quoted companies. We removed those firms belonging to the finance and insurance sectors. As a result, our final sample was made up of 117 corporations from different sectors.

Quoted companies are required by law to disclose relevant information to stockholders through their web sites. Furthermore, these corporations do business in a regulative environment in respect of corporate governance that is more demanding than that of the remaining Spanish firms. Such legal pressures aim at making Spanish

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companies more conscious of the ethical principles they must obey, by using the main Spanish corporations, leaders in their activity sectors, as models to follow. According to these arguments, it is likely that the remaining companies will adopt, to a greater or lesser extent, the behaviour of these pioneer companies in the disclosure of strategic information. Therefore, the analysis of these quoted companies would provide the remaining companies in their activity sectors with a model to follow.

Moreover, we selected this sample given that it is a set containing the largest Spanish companies and the most important ones in the Spanish Stock Market. In this sense, the largest corporations are more visible and more likely to have sufficient resources and incentives to adopt a policy of voluntary online disclosure, and therefore a lack of such a disclosure are likely to reflect a conscious choice.

Finally, the financial data necessary for the empirical analysis were obtained from the AMADEUS database for December 2005. AMADEUS database compiles financial company information and business intelligence for companies in across Europe and contains comprehensive information on over 14 million European companies.

*5.2 Creating a disclosure index*

After selecting the sample, we carried out an analysis of the sample companies' web sites. In order to perform this analysis we created a disclosure index. This type of index forms a part of content analysis and is one of the main techniques used to study the information disclosed by companies (Ortiz and Clavel, 2006). There is extensive accounting literature relating to the use of disclosure indices to measure the information contained in the annual reports of companies (Giner, 1997; Prencipe, 2004).

To create the index, we initially considered several descriptive studies that analyse the amount of voluntary information provided by companies on their web sites, in different countries, such as the US (Ettredge *et al.*, 2001), Germany (Marston and Polei, 2004), Austria (Pirchegger and Wagenhofer, 1999), Denmark (Petersen and Plenborg, 2006) and Spain (Larrán and Giner, 2002). Such studies focus on verifying a set of issues involving information disclosed on web sites, taking binary values (1: presence of the information sought; 0: absence of the information sought). Then, the values obtained are aggregated and, where appropriate, weighted.

In relation to the specific items about strategic information, we have considered the papers of Campbell and Beck (2004), Leuthesser and Kohli (1997), Campbell *et al.* (2001), Santema and Van de Rijt (2001), Bartkus *et al.* (2002) and Santema *et al.* (2005). Likewise, we have included several strategic factors according with several textbooks.

After defining the items in the index, the next stage was to quantify them. When using this methodology to find the levels of disclosed information for each item, a binary variable can be chosen, which takes a value of 1 or 0, depending on whether the data is reported (Cooke, 1989) or, alternatively, a score can be estimated between 1 and 0. Although the latter solution may be considered conceptually superior, it is typically a highly subjective evaluation (Giner, 1995). In this study, according to the approach most used in researching online disclosure, we have opted for binary variables.

Finally, another relevant issue is the possible weighting of the items, as in some studies (e.g. Pirchegger and Wagenhofer, 1999; Gandía, 2001). In our research, we have chosen an unweighted index, given that, according to Giner (1997), there is some arbitrariness inherent in the use of any weighted index. Moreover, studies that use both weighted and unweighted indices draw similar conclusions from both types of indices

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(Choi, 1973; Chow and Wong-Boren, 1987). As a result, we have chosen the aggregation of the scores obtained for each item in an unweighted index (as in Cooke, 1989; Raffournier, 1995; Giner, 1997).

*5.3 Explanatory model proposed*

In order to research the hypotheses established, we analysed factors potentially influencing the amount of information disclosed on the internet measured by the index defined above.

With that goal in mind, we propose the following Model (1), in which the amount of disclosed information on web sites is a function of the Board activity, its size, CEO duality, outside directors, shareholder representatives, corporate size, industry, profitability, leverage and ownership dispersion:

$$\begin{aligned} &\textit{Disclosed Information Online} = f\ (\textit{Board Activity},\ \textit{Board Size},\ \textit{CEO duality},\\ &\textit{Outside directors},\ \textit{Shareholder representatives},\ \textit{Corporate Size},\ \textit{Industry},\\ &\Pr\textit{ofitability},\ \textit{Leverage},\ \textit{Ownership Dispersion} \end{aligned} \tag{1}$$

Model (1) can be empirically estimated by using Equation (2):

$$\begin{aligned} SIDOL_i = {} & \beta_0 + \beta_1 ActBoard_i + \beta_2 SizeBoard_i + \beta_3 CEODual_i + \beta_4 OutsidDirect_i \\ & + \beta_5 Sharehrepr_i + \beta_6 MC_i + \sum \beta_{7k\rho K} + \beta_8 ROA_i + \beta_9 Lev_i \\ & + \beta_{10} OwnershipDispers_i + \varepsilon \end{aligned} \tag{2}$$

in which:

- $SIDOL_i$ = the disclosure index for strategic information obtained after analysing company $i$'s web site;
- $ActBoard_i$ = the number of meetings of the Board for company $i$ during the year 2005;
- $SizeBoard_i$ = the number of members of the Board for company $i$, as a proxy for the size of that Board;
- $CEODual_i$ = a dummy variable that takes the value 1 if the Chairperson of the Board holds the managerial position of CEO and the value 0, otherwise;
- $OutsidDirect_i$ = the percentage of outside Board members sitting on the Board;
- $Sharehrepr_i$ = a dummy variable that takes a value of 1 when outside directors sit on the Board, and otherwise has a value of 0;
- $MC_i$ = company $i$'s market capitalisation as a variable related to corporate size;
- $\Sigma\beta_{7k}\rho_K$ = are dummy variables that take a value of 1 if the company belongs to industry $k$ and 0, otherwise; the sectors we have

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considered are: Services, Transportation, Industry, Energy and Construction, according to the industry classification established by the CNMV;

$ROA_i$ = the return on assets for company $i$, defined as the ratio between operating income and total assets;

$Lev_i$ = company $i$'s leverage, established as the ratio between debt volume and total assets;

$OwnershipDispers_i$ = the percentage of company $i$'s shares that are not owned by blockholders and by the executives and directors, as a measure of free float.

Model (2) was checked empirically through a linear regression, estimated by OLS. As mentioned above, the dependent variable was obtained by analysing the items in the disclosure index on the web sites. The independent variables were taken from the AMADEUS Database and the companies' reports.

Furthemore, we must point out that Lynk *et al.* (2008) have found that Board size and Board independence are positively explained by company size and debt. Ning *et al.* (2007) have evidenced that Board size differs significantly across industries and the cross-sectional variation in Board size is driven by various economic forces such as profitability. Similarly, Boone *et al.* (2007) have observed a significant relation between Board size and Board independence.

Therefore, in this paper we have run a regression of BOARDSIZE, BOARDINDEP and BOARDACTIVITY on FIRMSIZE, LEVERAGE, ROA and Industry Dummies. Then we used the residuals from these regressions as a substitute for BOARDSIZE, BOARDINDEP and BOARDACTIVITY. By doing so, we account for the marginal effects of Board size, independence and activity on Board parameters, while avoiding any problems associated with multicollinearity.

## 6. Empirical results

### *6.1 Statistical description*

From the data shown in Table I, we find that Spanish companies, on average, give out little strategic information; more specifically, only two out of the eight items analysed are disclosed. However, the maximum statistic indicates that some firms reveal information on seven out of the eight items.

In Table II, we summarise the frequencies and their percentages for each item analysed. We detected that the item Objectives, mission and company philosophy is the one most frequently disclosed (79.50 per cent), showing an important gap with the three subsequent items, which display a percentage frequency lower than 40 per cent. The second item is "Strategic position of company in its sector", disclosed by 37.6 per cent of the companies analysed. Following these, the items "Company strategic planning" (29.9 per cent) and "Information on production processes" (27.4 per cent) are

**Table I.** Disclosure index: descriptive statistics

| Index | Mean | Minimum | Maximum | Standard deviation |
|---|---|---|---|---|
| Strategic information | 2.0855 | 0 | 7 | 1.50042 |

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| Items | | Frequency | Percentage |
|---|---|---|---|
| 1 | Objectives, mission and company's philosophy | 93 | 79.50 |
| 2 | Strategic alliances | 10 | 8.50 |
| 3 | Strategic position of company in its sector (leader, 2nd) | 44 | 37.60 |
| 4 | Company strategic planning (projects of expansion into other markets, products, regions) | 35 | 29.90 |
| 5 | Company annual planning | 9 | 7.70 |
| 6 | Description of the competition context | 12 | 10.30 |
| 7 | Information on risks (financial, commercial, technical) | 9 | 7.70 |
| 8 | Information on production processes | 32 | 27.40 |

**Table II.**
Items of the disclosure index: frequency and percentage

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the ones most commonly disclosed. At the opposite end of the spectrum, the least disclosed items are related to "Company annual planning" and "Information on risks" with a frequency percentage of 7.7 per cent; "Strategic alliances" (8.5 per cent) and "Description of the competition context" (10.30 per cent).

The bivariate correlations between the variables analysed are summarised in Table III. None of the independent variables show a significant correlation with the volume of strategic information disclosed by companies.

*6.2 Multivariate analysis*

The results obtained after estimating the proposed models are shown in Table IV. The overall significance of the model ($R^2$) reaches 23.10 per cent at a confidence level of 95 per cent ($0.01 < p\text{-value} < 0.05$). Our findings are similar for both the initial model (column 2) and the model in which we have corrected multicollinearity problems (column 3).

In the last three columns we synthesise the results of the models estimated for MEETING, MEMBERS and OUTSIDE DIRECTORS variables. We can see that Board activity is marginally explained by outside directors. Board size is positively explained by Board independence, firm size and leverage level, and negatively associated[7] with the industry in which the company operates. Board independence displays a positive relationship with Board activity and Board size and a negative relationship with firm level of leverage.

With regard to the variables that could explain the disclosure of strategic information, four out of the 14 are statistically significant. In particular, ACTIVITY and TRANSPORTS show a negative and significant impact at a confidence level of 95 per cent. LEVERAGE, at the same confidence level, exhibits a positive effect. The coefficient of CEO DUALITY, significant at 0.10, indicates a positive effect on the disclosure of strategic information.

These findings would support *H1*, with negative effect, a sign opposite to what was expected according to Agency Theory postulates, as well as the positive sign expected for leverage. In part, we support *H3* (with a negative sign), since the effect is only linked to the existence of CEO DUALITY. Concerning the industry variables, the impact is only significant for the Transportation industry.

The variables ROA and SHAREHOLDER REPRESENTATIVES have a negative but non-significant effect on the estimated model. The remaining industry variables

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**Table III.**
Correlation Matrix
(Pearson correlations coefficients)

| | Index for Strateg Inform | Meetings | Members | CEO dual | Outside directors | Sharehold's representat | Ownersh dispers | ROA | Leverage | Corporat size | Services | Transpts | Industry | Energy | Construct |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Index for strategic inform | | | | | | | | | | | | | | | |
| Meetings | −0.102 | | | | | | | | | | | | | | |
| Members | 0.138 | 0.276 | | | | | | | | | | | | | |
| CEO duality | 0.188 | −0.049 | −0.021 | | | | | | | | | | | | |
| Outside Directors | 0.037 | 0.263 | 0.382 | −0.220 | | | | | | | | | | | |
| Shareholder representatives | −0.012 | 0.022 | 0.105 | 0.126 | 0.139 | | | | | | | | | | |
| Ownership dispersion | 0.117 | 0.320 | 0.068 | 0.000 | 0.157 | −0.163 | | | | | | | | | |
| ROA | −0.057 | −0.011 | 0.043 | −0.134 | −0.022 | 0.074 | −0.023 | | | | | | | | |
| Leverage | 0.072 | 0.054 | 0.283 | −0.050 | −0.026 | 0.095 | −0.084 | −0.027 | | | | | | | |
| Corporate size | 0.147 | 0.121 | 0.326 | 0.068 | −0.010 | 0.089 | 0.126 | −0.026 | 0.153 | | | | | | |
| Services | −0.047 | −0.105 | −0.085 | −0.057 | −0.089 | 0.036 | −0.200 | −0.016 | 0.174 | 0.009 | | | | | |
| Transportation | −0.176 | 0.157 | 0.151 | −0.072 | 0.057 | 0.047 | 0.136 | −0.001 | 0.043 | 0.328 | 0.326 | | | | |
| Industry | −0.101 | −0.072 | −0.276 | −0.009 | −0.028 | −0.040 | 0.055 | 0.050 | −0.177 | −0.085 | −0.582 | −0.190 | | | |
| Energy | 0.123 | 0.234 | 0.265 | 0.145 | 0.061 | −0.081 | 0.209 | −0.108 | −0.040 | 0.047 | −0.263 | −0.086 | −0.306 | | |
| Construction | 0.075 | 0.005 | 0.143 | 0.030 | 0.104 | 0.069 | 0.044 | 0.100 | 0.008 | 0.093 | −0.237 | −0.077 | −0.276 | −0.124 | |

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| | MODEL Dependent variable Disclosure index for strategic information | | | | MODEL Dependent variable Meetings | | Members | | Outside Directors | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Standardised | | Standardised | | Standardised | | Standardised | | Standardised | |
| | $\beta$ | $t$ | $\beta$ | $t$ | $\beta$ | $t$ | $\beta$ | $t$ | $\beta$ | $t$ |
| Intercept | – | 0.542 | – | 1.406 | – | 0.795 | – | 2.269** | – | 4.952*** |
| Residuals | | | | | | | | | | |
| Meetings (ACTIVITY) | −0.229 | −2,013** | −0.177 | −1.638** | – | – | 0.063 | 0.712 | 0.190 | 1.832* |
| Residuals | | | | | | | | | | |
| Members | 0.068 | 0,489 | 0.087 | 0.769 | 0.099 | 0.712) | – | – | 0.486 | 4.035*** |
| CEO duality | 0.210 | 1,883* | 0.210 | 1.883* | – | – | – | – | – | – |
| Residuals | | | | | | | | | | |
| Outside Directors | 0.143 | 1,155 | 0.124 | 1.019 | 0.212 | 1.832* | 0.348 | 4.035*** | – | – |
| Shareholders' representat. | −0.061 | −0,558 | −0.061 | −0.558 | – | – | – | – | – | – |
| Ownership dispersion | 0.138 | 1,192 | 0.138 | 1.192 | – | – | – | – | – | – |
| ROA | −0.072 | −0,684 | −0.070 | −0.666 | 0.003 | 0.032 | 0.089 | 1.072 | −0.051 | −0.514 |
| Leverage | 0.226 | 2,010** | 0.190 | 1.796** | 0.159 | 1.430 | 0.214 | 2.458** | −0.216 | −2.080** |
| Corporate size | 0.163 | 1,404 | 0.169 | 1.529 | 0.001 | 0.008 | 0.254 | 2.846*** | −0.168 | −1.542 |
| Services | 0.152 | 0,511 | 0.095 | 0.339 | 0.036 | 0.119 | −0.677 | −2.996*** | 0.214 | 0.762 |
| Transportation | −0.240 | −1,990** | −0.264 | −2.219** | 0.182 | 1.558 | 0.066 | 0.693 | 0.023 | 0.205 |
| Industry | 0.106 | 0,342 | 0.030 | 0.103 | 0.137 | 0.439 | −0.744 | −3.168*** | 0.273 | 0.932 |
| Energy | 0.131 | 0,611 | 0.073 | 0.345 | 0.217 | 1.025 | −0.211 | −1.253 | 0.065 | 0.322 |
| Construction | 0.102 | 0,487 | 0.077 | 0.380 | 0.072 | 0.344 | −0.331 | −2.027** | 0.202 | 1.027 |
| $R^2$ | 0.231 | | 0.231 | | 0.168 | | 0.468 | | 0.256 | |
| $F$ | 1.626** | | 1.626** | | 1.619 | | 7.038*** | | 2.756*** | |

**Note:** *$p$-value < 0.01; **$p$-value < 0.05; ***$p$-value < 0.01

**Table IV.**
Multivariate analysis results: multiple linear regression

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(Services, Industry, Energy and Construction), as well as the variables MEMBERS, OWNERSHIP DISPERSION and CORPORATE SIZE, display a positive but non-significant effect.

The results of the variable ROA are consistent with several authors such as Larrán and Giner (2002), Oyelere *et al.* (2003), Giner *et al.* (2003), Marston and Polei (2004), Prencipe (2004) and Magness (2006). The absence of significant results for CORPORATE SIZE has been previously evidenced by Khanna *et al.* (2004) and Ortiz and Clavel (2006) for European multinationals.

As for Board size (MEMBERS), our findings show the positive effect also obtained by Pearce and Zahra (1992) and Dalton *et al.* (1999), who found that the number of directors was positively related to the processes for planning new strategies; however, it is not statistically significant in our analysis. In relation to ownership structure, the absence of relevant econometric results increases the empirical evidence, stressed by McKinnon and Dalimunthe (1993), Saada (1998), Leuz (1999) and Prencipe (2004).

But these last two insignificant results may be derived from the typology of strategy information that this paper has considered. Hence, they may suggest that Boards of Directors and blockholders are not specially prone to the disclosure of information that may generate adverse effects for the company, either negative reactions in stock prices (in the case of negative information) or competitive disadvantages (in the case of positive information).

Beyond Agency Theory, we have observed that Spanish companies in which the Chairperson of the Board is the same person as the CEO and in which there is a lower frequency of meetings disclose a greater amount of strategic information. A more in-depth study of these findings may suggest that a high number of meetings may not be the appropriate indicator of the fact that the Board works with adequate depth and breadth, but is operating in a managerial fashion, exceeding its function (Vafeas, 1999).

It is a quite unexpected finding for the Spanish case since several authors argue that more meeting frequency could reduce the asymmetric information problems. One possible explanation would have to do with the fact that Spanish Boards are dominated by executives, who are the main individuals responsible for the company"s strategy (Finkelstein and Hambrick, 1996). Consequently, they may not include the firms' strategic decision-making or even the disclosure practices in the meeting agenda.

With regard to the effect of the Board's independence, the results show the lack of impact regarding the composition, whereas CEO Duality remains positive. This finding may be due to the fact that this duality guarantees the unity of management inside the corporation (Zahra *et al.*, 2000) and a great knowledge of the firm's operating environment (Finkelstein and D'Aveni, 1994). On the other hand, outside directors are usually part-time, which means it is more difficult for them to understand the complexities of the firm (Baysinger and Hoskisson, 1990). This situation causes different levels of participation in the decision-making processes (for example, in the decisions on information disclosure). Similar results were obtained by Forker (1992), Ho and Wong (2001), Eng and Mak (2003), Haniffa and Cooke (2005), Boesso and Kumar (2007) and Lim *et al.* (2007), who found a negative or non-significant relationship between information disclosure and the presence of independent directors on the Board. But these situations could be more serious in the Spanish context since non-executive directors handle a lower volume of information in relation to the most important decisions concerning the company's management (Melle, 1999).

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However, it is also necessary to point out that outside directors may protect stockholders' interests as opposed to other stakeholders like competitors, who could be interested in strategic information (Lim *et al.*, 2007). Along this line, executives would disclose more future information to protect their wealth, to increase their reputation, etc. although it could generate several costs for shareholders.

Leverage is one of the factors associated with a greater amount of strategic information disclosure, showing the companies' future prospects and likelihood of success. Such disclosure can help reduce agency costs and the potential conflicts between owners and creditors. These findings are in line with those of Higgings and Diffenbach (1985), who verified that this kind of information is widely used by professional investors in the evaluation of firms in financial distress.

With regard to the industry, we have shown that the companies in the Transportation sector may adopt similar patterns regarding the strategic information they disclose, so as to keep investors from interpreting the lack of information as an adverse sign. In this sector, firms choose to reveal a lower level of strategic information on their web sites. According to previous works, this lower disclosure rate may also be justified on the grounds of confidentiality (Vacas *et al.*, 2005) and the fear of possible lawsuits (Santema and Van de Rijt, 2001).

Finally, despite the reasonableness of the assumptions in disclosure theories, we have not found any statistically significant relationship between voluntary disclosure on the internet and corporate size and profitability, as other works have (e.g. Oyelere *et al.*, 2003).

## 7. Conclusions

The voluntary disclosure of strategic information is an increasingly common corporate practice for several reasons. For instance, it allows the company to stand out from others and impacts positively on the evaluation of the corporation by finance professionals. In general, the studies performed in other countries have shown that the disclosure of strategic information focuses on revealing the firm's global mission and corporate strategy, and the fulfilment of objectives in prior years; less frequently, action plans are included as well.

The reasons that can explain the voluntary disclosure of future information may be very different from other types of reports owing to the competitive disadvantage that could arise if competitors use this data and/or the firms do not achieve the objectives pursued. Moreover, this information will not always be positive; hence, it can lead to a decline in stock prices or in the firm's leverage capacity.

In order to observe the level of strategic information disclosed in the Spanish context, we have created an index that includes several items analysed in previous studies and that we have complemented with the contents on this topic from several textbooks. The index was quantified as a binary variable, which takes a value of 1 or 0, depending on whether the data is reported or not.

By analysing a new commonly used channel for disseminating corporate information –the internet– this study has found that these companies mainly disclose information about their objectives, mission or firm philosophy on their web sites. Less frequently, they reveal their strategic position in their respective sectors, their strategic plans and information about their production processes. In contrast,

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data about which firms disclose less information are associated with the annual plan, risks, strategic alliances and the description of the competitive context.

We subsequently analysed the explanatory effect that several factors have on the disclosure of voluntary strategic information, through a statistical dependence model – specifically, multivariate linear regressions. To do so, we adopted an Agency Theory framework, although we have included several reflections on the priorities of the responsibilities in the Board. On the one hand, directors have to reduce information asymmetry problems in order to control the executives' actions effectively. On the other hand, outsiders must protect shareholders' interests from proprietary costs (costs associated with the drawing-up of information, litigation risks or negative reactions to bad news in stock prices) that are linked to future data.

In order to alleviate multicollinearity problems, we accounted for the marginal effects of Board size, independence and activity on auditing committee effectiveness.

Our results show that the most leveraged organisations reveal more information about future projects and their success probabilities, possibly owing to diminishing the agency costs and the conflicts of interest with debtors and to satisfying the demand for information on behalf of investors.

With regard to the effect of corporate governance, we have found that companies in which the Chairperson of the Board is the same person as the CEO and in which there is a lower frequency of meetings – less active Boards – disclose a greater amount of strategic information. The positive role of CEO duality and the insignificant effect of non-executive directors could be justified by the stakeholder model hypothesis, in which managers are not opportunistic agents but rather moral individuals and their role is seen as achieving a balance between the interests of all stakeholders (Shankman, 1999, p. 322).

Jointly considered, these findings seem to indicate an important participation of upper-level management in the decision to disclose strategic information online in order to satisfy creditor demands and public scrutiny, except in the Transportation industry, which seems to show some opposition to the disclosure of strategic data.

These results reinforce those of Babio *et al.* (2003), who, through interviews, point out that companies disclose a greater amount of voluntary information when the benefits to investors outweigh the costs associated with disclosing to competitors, and when it involves benefits that are related to capital markets – such as increases in stock prices or better access to markets. Moreover, these results could be affected by the characteristics of the Spanish context, such as the strong presence and power of executives who can make decisions without taking independent directors into consideration. In this sense, according to Zahra (1990)'s classification, Spanish Boards would be included in the first level – the less active one – representing and protecting stockholders' interests, without becoming involved in the development and implementation of corporate strategies.

As for previous empirical evidence, we have observed that large shareholders do not participate in the decision of disclosing higher volumes of strategic information. These findings are in accordance with those obtained in other countries, such as France and Australia, by Depoers (2000) and Lim *et al.* (2007). However, unlike Lim *et al.* (2007)'s results, we have detected that, in Spain (a context featured by a high presence of executives on Boards), the outside directors do not play the r' opinions are the key explanatory factors for voluntary strategic disclosure.

As limitations of the current study, we take the volume of information disclosed as a proxy for substantive disclosure. Even though the "level of disclosure" is measured by the number of indicators (e.g. quantitative measure), these indicators should also be qualitative in order to be informative. However, our findings would be more robust if a more in-depth study of qualitative features was undertaken.

Finally, examining the relation between voluntary disclosure of strategic information and corporate governance has many other future avenues for research. From an international perspective, differences in legislation and institutional factors and their impact on disclosure quality could be addressed. It would thus be of interest to extend the current work to different countries, where the disclosure of strategic information may be voluntary or compulsory.

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## Notes

1. In this article the disclosure of strategic information refers to information about the future actions of firms. This term is very different from the strategic disclosure of information, which refers to the disclosure of information that firms consider useful strategically. Under Circular 1/2004 of the Comisión Nacional del Mercado de Valores (*CNMV*, 2004; Spain's equivalent of the US's SEC), which requires the disclosure of relevant information on web sites, the term 'strategy data' is not clearly specified.
2. It is necessary to emphasize that not all the information voluntarily disclosed by firms has a positive effect on shareholders' decision-making. Thus, Gelb and Zarowin (2002) found that enhanced annual report disclosures do not make current stock prices more informative. In this line, Banghoj and Plenborg (2008) evidenced that, in fact, a higher level of voluntary disclosure reduces the association between current returns and future earnings, owing to the limited usefulness of the information. Several authors, such as Cianci and Falsetta (2008), have observed that relevant information for investors is associated with negative and positive future company actions.
3. The disclosure of strategic information initially focused on revealing firms' general missions (Bart, 1997; Leuthesser and Kohli, 1997; Campbell *et al.*, 2001), but currently its content has been expanded to include the corporate strategy and the fulfilment of goals from prior years, and, less frequently, action plans (Santema and Van de Rijt, 2001). Furthermore, there has also been a change in the mechanism or channel used for disclosure. It has evolved from traditional annual reports to the inclusion of such information to corporate web sites (Bartkus *et al.*, 2001, 2002). This process may be due to the importance of the internet as a mechanism for revealing corporate reputation (Campbell and Beck, 2004).
4. This lower development of the financial markets and the stability and concentration of stock ownership suggest that the Spanish corporate governance system is an internal one based on the Board and the supervisory role of large shareholders (Fernández and Arrondo, 2005).
5. However, the obtaining of a lower cost of capital through voluntary disclosure is controversial in previous literature. For instance, while Francis *et al.* (2005) and Francis *et al.* (2008) find that more voluntary disclosure is associated with a lower cost of capital, neither Botosan (1997) nor Wang *et al.* (2008) obtain evidence that companies benefit from extensive voluntary disclosure by having a lower cost of capital.
6. Both the rapid adoption of the internet as a way to disclose corporate information and the heterogeneity of the content published online have led to a need for regulation, through a standardization process in the national and international contexts. The objective of such a process would be to harmonise contents and formats to make online information comparable. Among these norms, several are notable: within the international context, the

report issued by the International Federation of Accountants (IFAC, 2002); within a regional context, the European Commission Directive issued in 2004 deals with the harmonisation of transparency requirements and considers the internet to be an appropriate way to disclose financial information, and article 17 allows the use of electronic methods for revealing information to stockholders; within the national context of this study – Spain – there is one compulsory rule, Circular 1/2004, of the Comisión Nacional del Mercado de Valores (*CNMV*, 2004), establishing the type and volume of information whose inclusion is required on the web sites of quoted corporations.

7. The Services, Industry and Construction Sectors present a negative relationship with Board size. In other words, in these sectors, the companies' Board size is lower than that in Energy or Transportation industries.

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## Corresponding author

Isabel-María García Sánchez can be contacted at: lajefa@usal.es

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